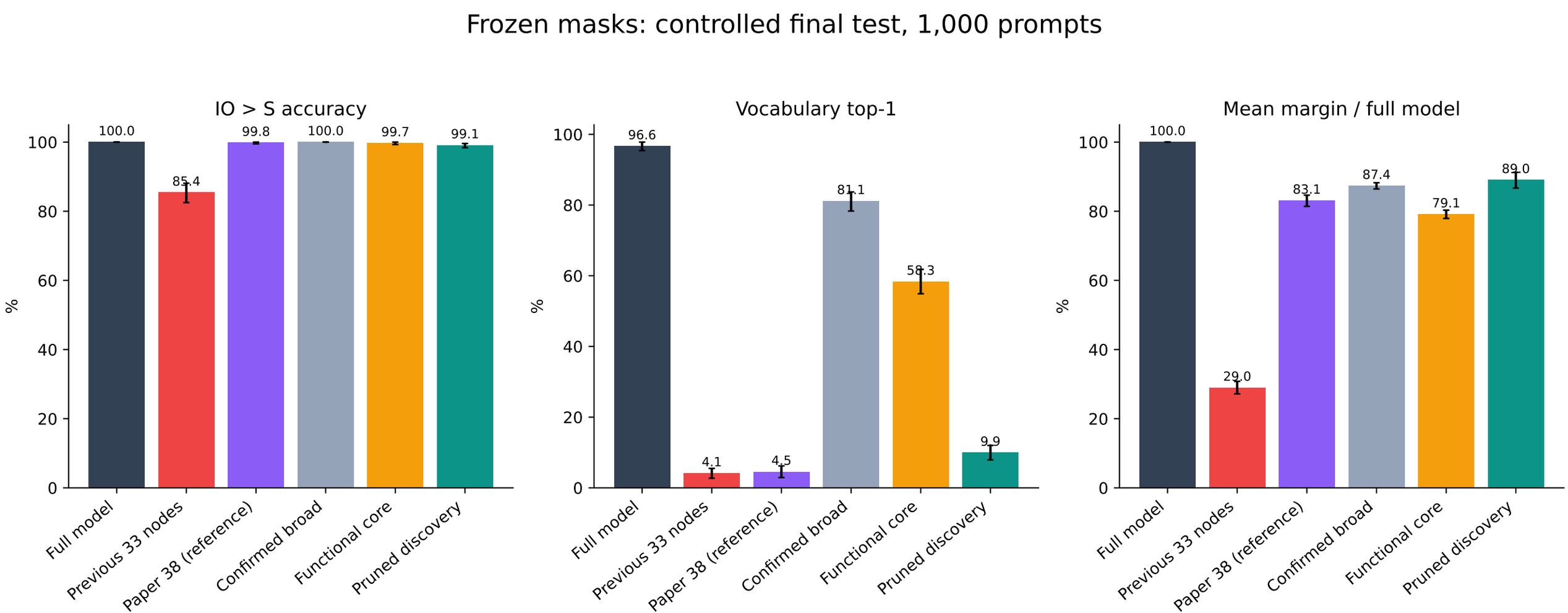
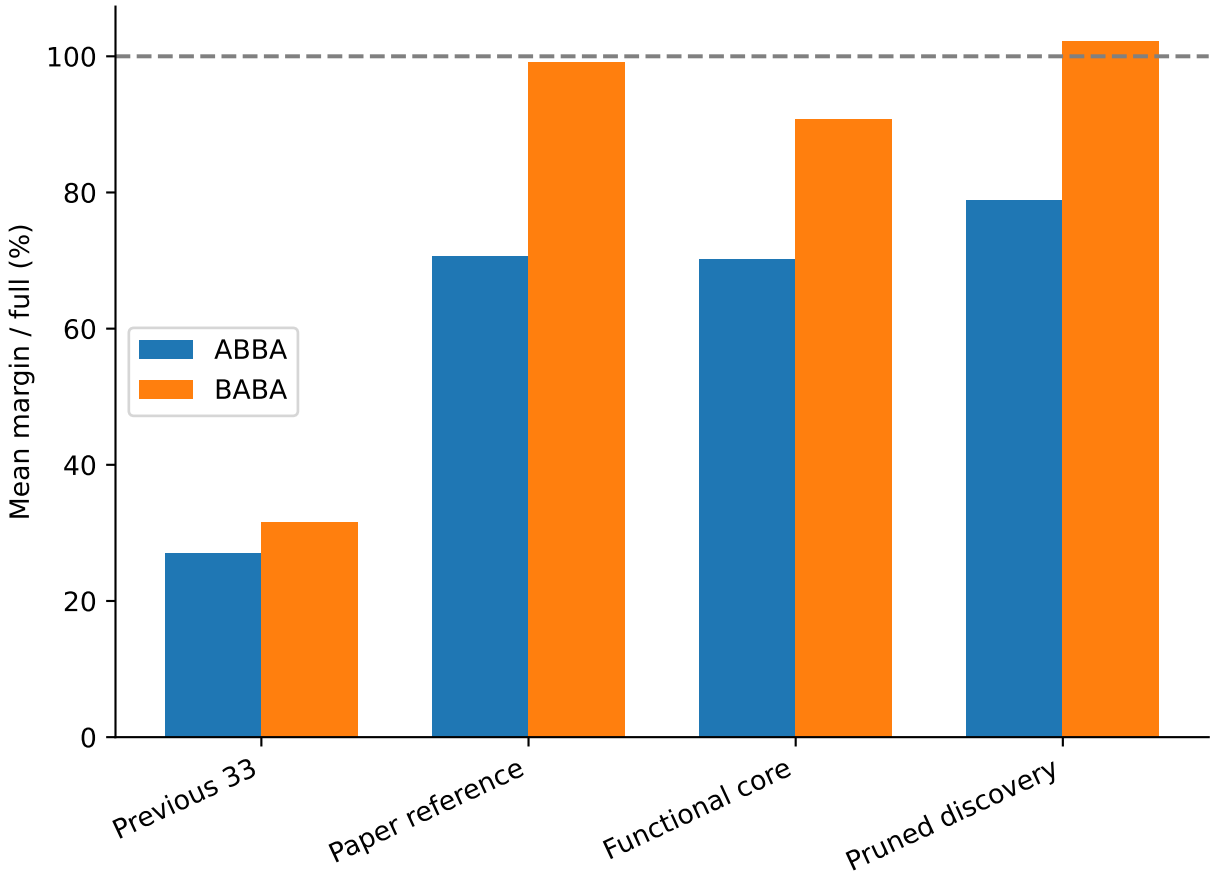
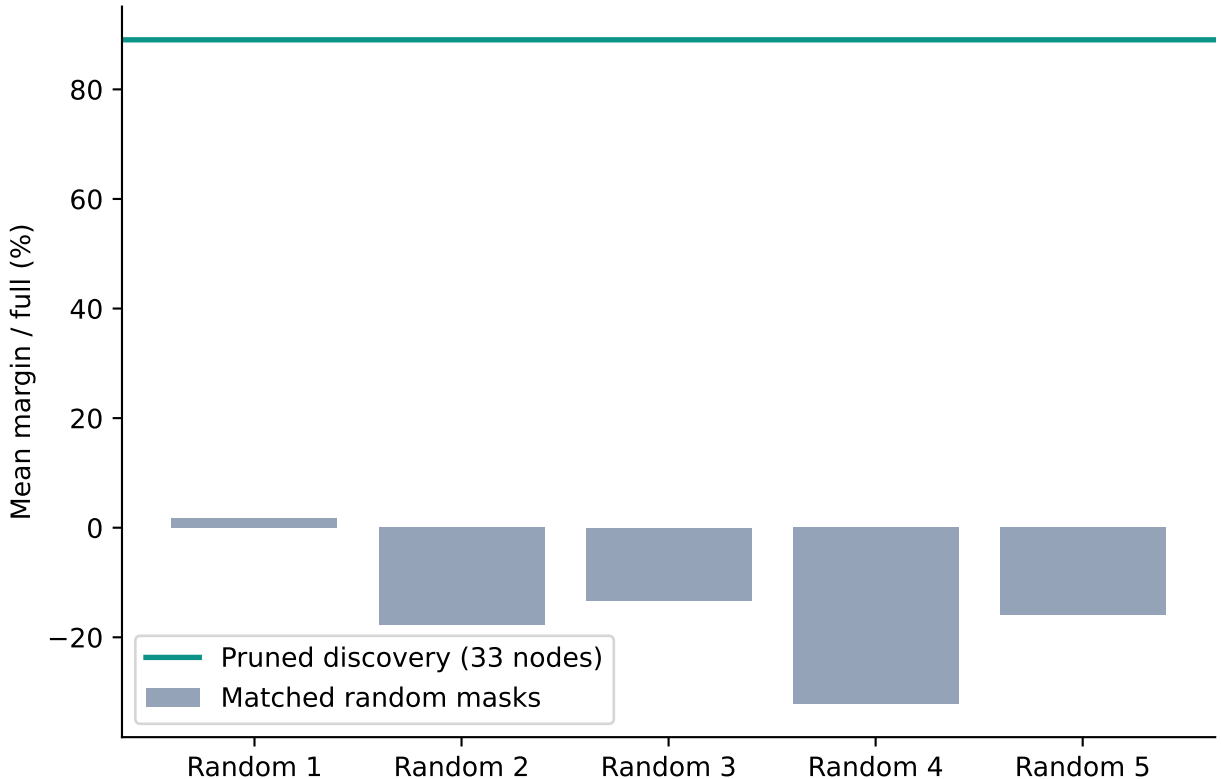



Same test set and 3,000 independent ABC reference prompts. Embeddings and MLPs retained. Error bars: 95% matched-context cluster bootstrap, stratified by template. Zero-error bars are degenerate. Paper mask uses inferred positions.

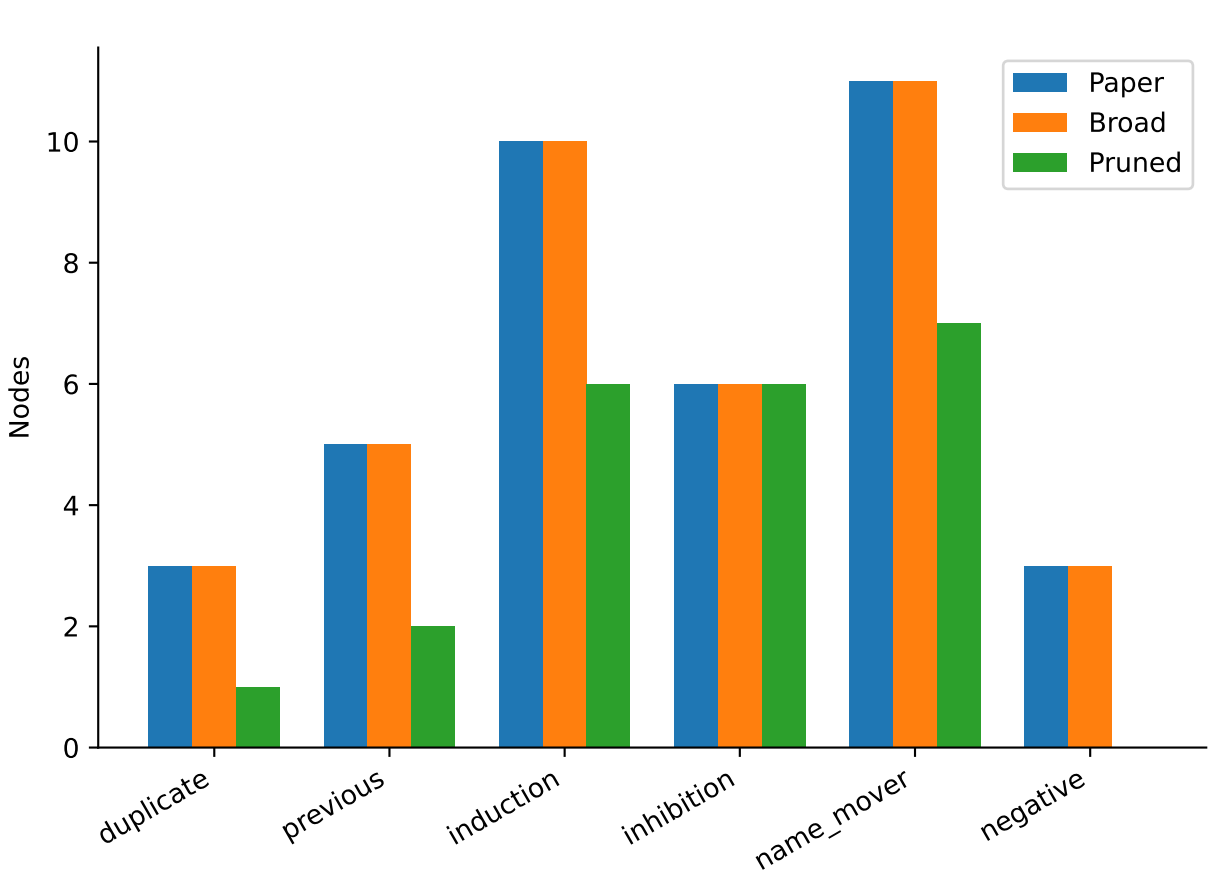
# Ordering cancellation and matched random controls

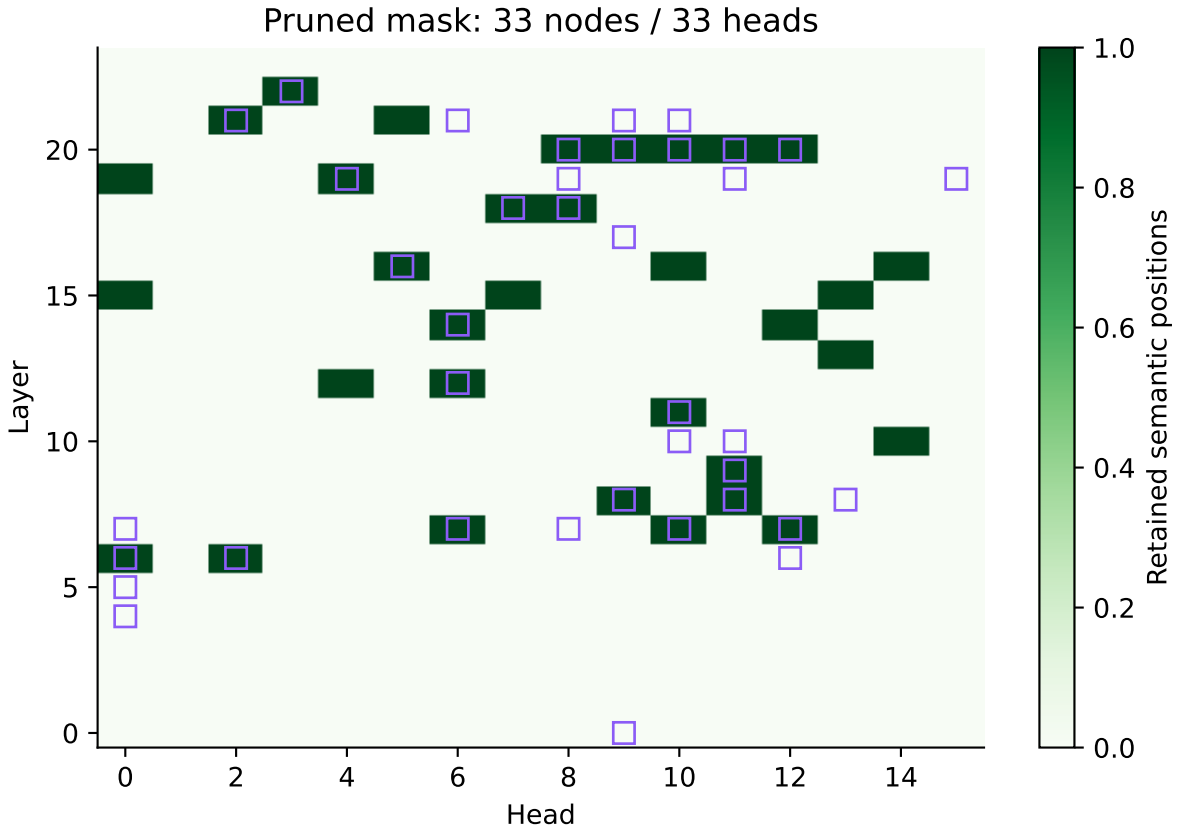


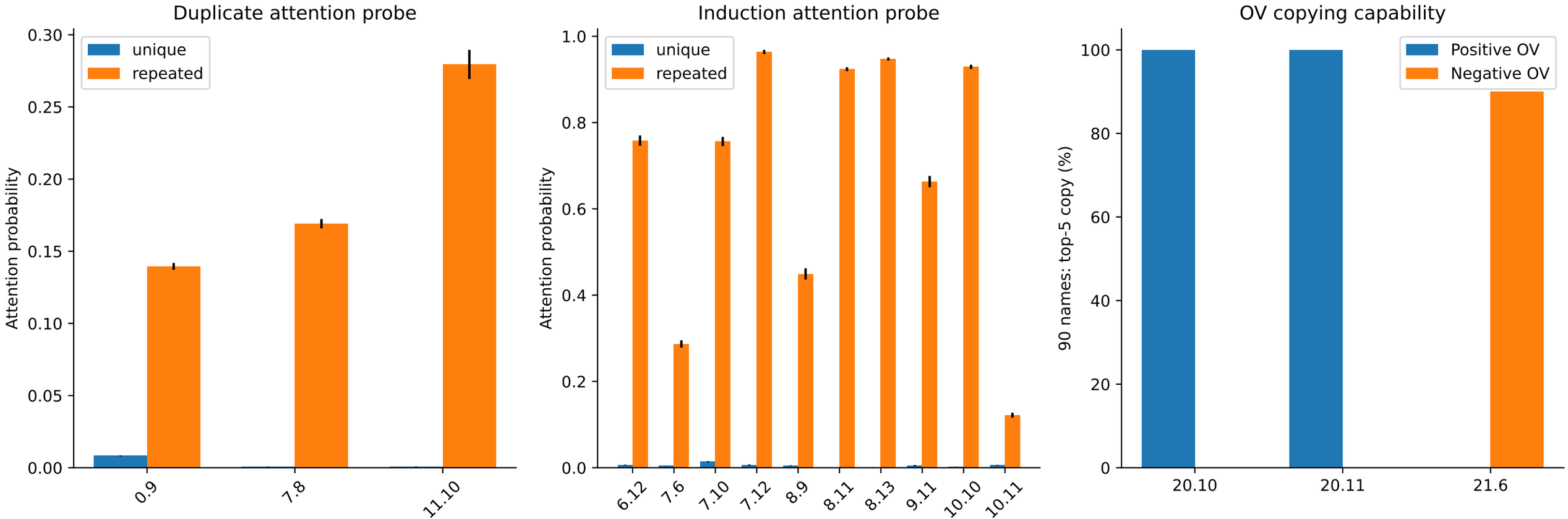

Five random masks match the primary circuit node counts at each layer and semantic role. Finite controls, not a p-value. Ratios above 100% are possible and are not clipped.

# Independent discovery versus published functional families



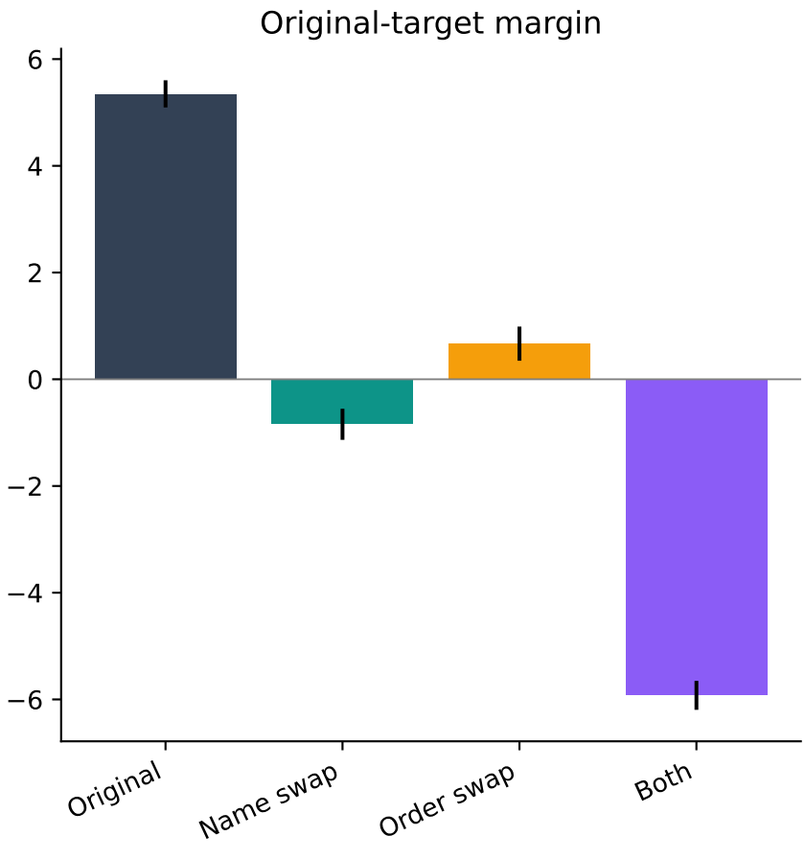



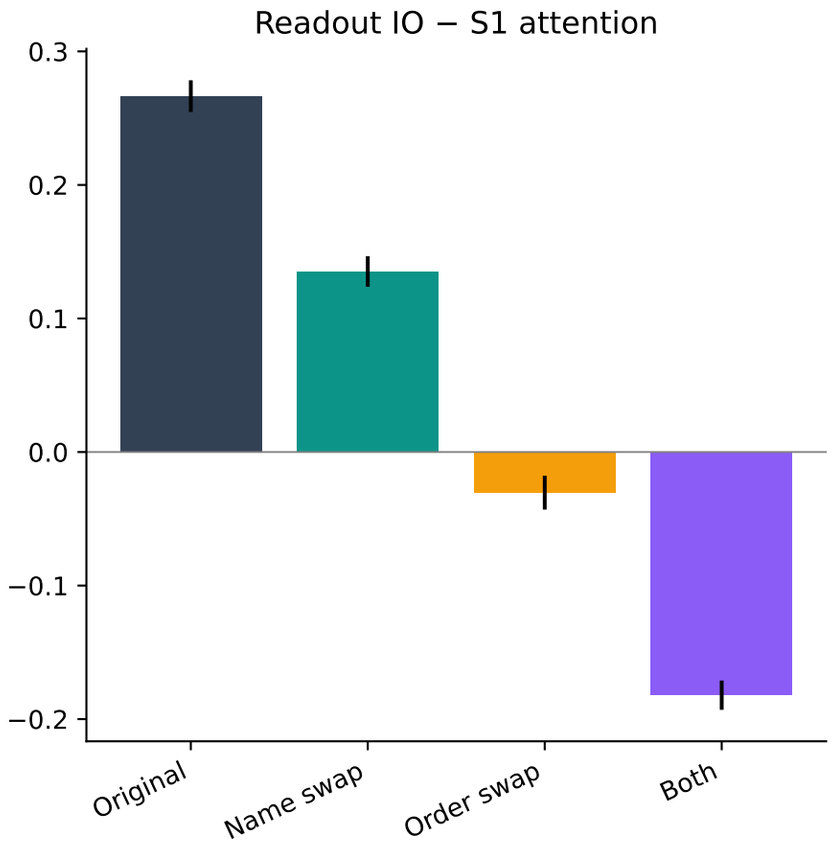

Purple squares mark paper heads; green intensity counts retained positions. Overlap is descriptive. Pruning used development performance, not published head IDs. A shared head is not proof of the same functional role.

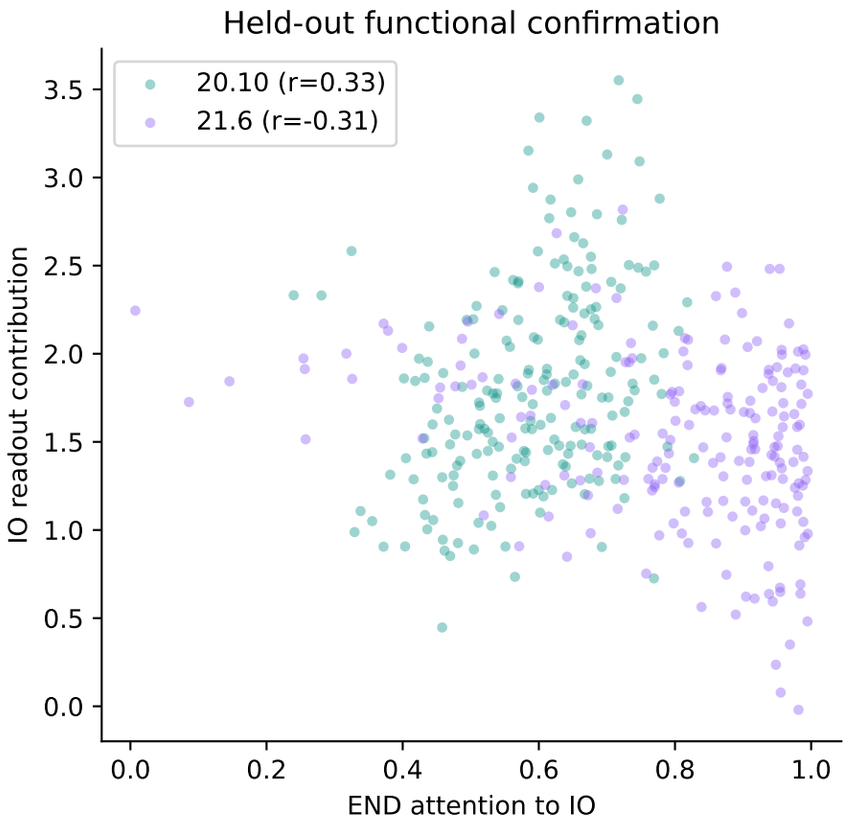



Random-token probes: 64 repeated and 64 unique sequences, adapted to BLOOM. 95% normal error bars. Selected published heads shown for comparison only. OV probes test capability; attention patterns alone do not identify a causal circuit.

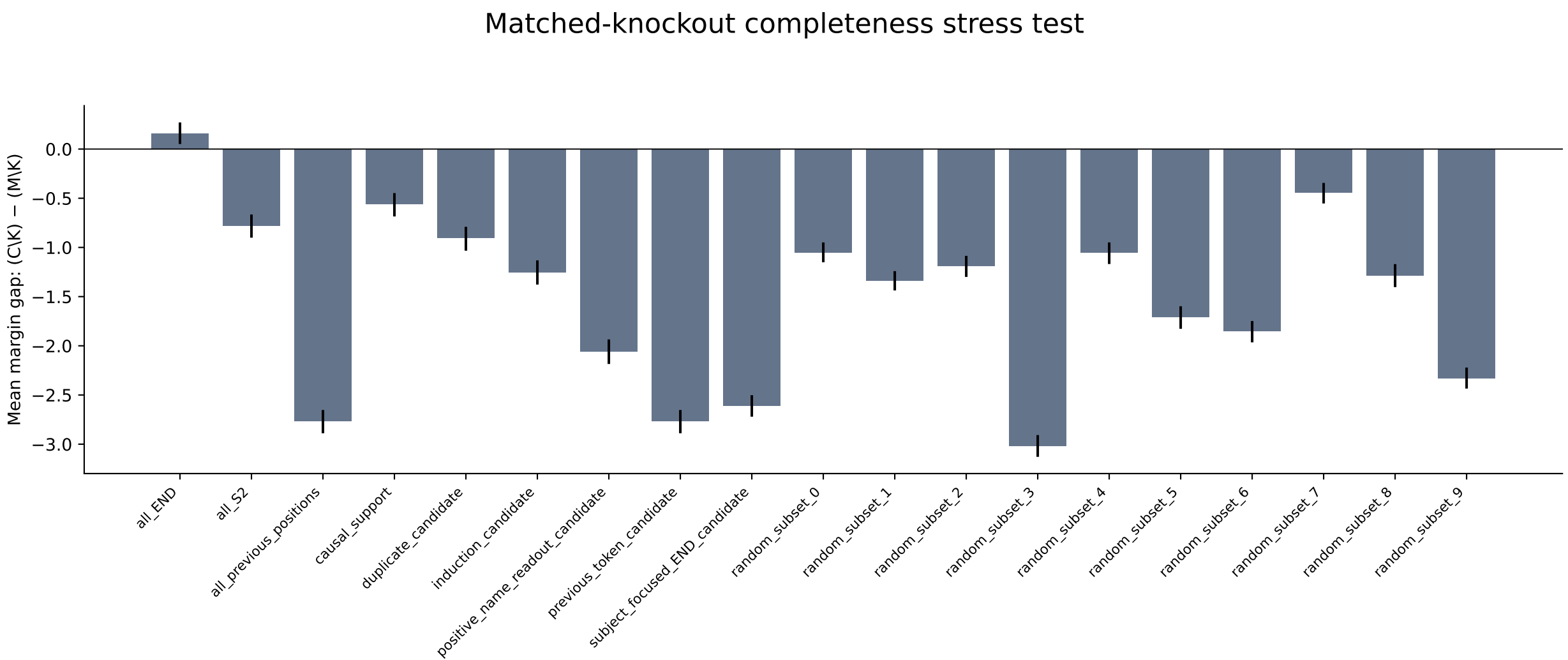
# Causal sensitivity to names and order



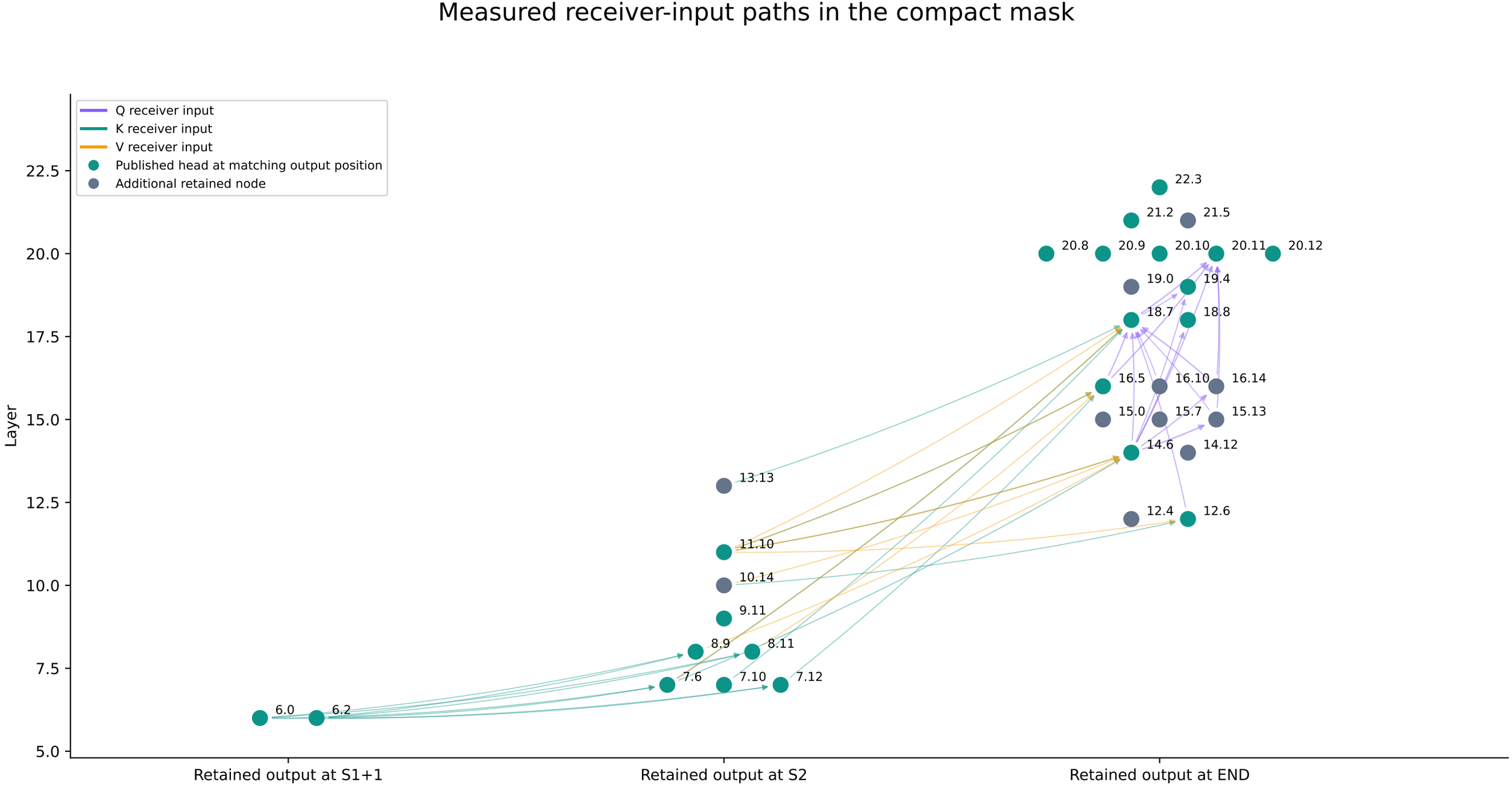




200 development prompts. Joint END-output replacement in subject-focused heads, scored on original IO/S targets. BLOOM uses ALiBi: this tests token/order dependence, not learned-position embedding copying. Scatter correlation is not causal proof.




1000 test prompts. Same knockout in circuit C and full model M. Nonzero gaps expose unmodeled compensation. Finite group/random tests do not prove completeness or conditional minimality. Error bars: 95% template-stratified context-cluster bootstrap.




Shown: 40 of 294 supported endpoint paths (up to eight strongest per receiver route), confirmed on 200 separate prompts. Color: Q/K/V input. Induction K is patched at S1+1; subject K/V at S2; Q at END. Output masks differ from input positions. This is a partial graph, not a completeness claim.